GOLD STANDARD ROADMAP: BIBLIOMETRICS, SCIENTOMETRICS & TECH-MINING TOOLS

Integrated Feature Inventory for Top-Tier Journal Manuscript Production

Sources: Original research inventory (2010–2025 literature), Gold Standard phased implementation analysis, and competitive-advantage platform analysis (Horizon 1–4 framework). Targets: Nature, Scientometrics, Journal of Informetrics, Technological Forecasting and Social Change, Journal of Business Research, Research Policy.

Tools referenced: Bibliometrix/Biblioshiny, VOSviewer, CiteSpace, SciMAT, ScientoPy, PyBibX, Gephi, Pajek, HistCite, CitNetExplorer, Sci2, BibExcel, Publish or Perish, VantagePoint, OpenRefine, Dimensions, OpenAlex, Lens.org, PatentInspector, PatSeer, AcclaimIP, Minesoft Origin, Scite.ai, Semantic Scholar, ResearchRabbit, Open Knowledge Maps, Rayyan, SciVal, InCites, Linguamatics I2E, Biblio AI (bibliometrix 5.0)

READING GUIDE

[PHASE n]	Implementation phase from Gold Standard analysis (1–4)
[H n]	Horizon from competitive-advantage analysis (1–4)
[GAP]	Not fully implemented in any major open-source tool
[NEW-GS]	Introduced by Gold Standard phased analysis
[NEW-H]	Introduced by Horizon competitive-advantage analysis

ARCHITECTURE: The roadmap is organized into five parts.

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PART A. STUDY DESIGN & CORPUS CONSTRUCTION    (Phase 1 / Horizon 1)
PART B. TEMPORAL & FRONTIER DYNAMICS          (Phase 2 / Horizon 2)
PART C. STRUCTURAL ANALYSIS & NOVELTY MINING  (Phase 3 / Horizon 2–3)
PART D. AI-DRIVEN INTERPRETATION & AUTHORING  (Phase 4 / Horizon 3–4)
PART E. EXTENDED DOMAIN FEATURES              (Performance, Visualization,
        Reproducibility, Tech Mining, Patent Analysis)
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COMPETITIVE ADVANTAGE STACK (summary):

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Rayyan-like rigor        →  corpus construction (Part A)
PRISMA-grade reporting   →  transparency and reproducibility (Part A, Part E)
VOSviewer/SciMAT/
CitNetExplorer depth     →  science mapping (Part B, Part C)
Scite/Lens/Dimensions    →  evidence linkage and novelty (Part C, Part D)
Open Knowledge Maps /
ResearchRabbit discovery →  gap finding (Part C)
Biblio AI interpretation →  manuscript production (Part D)
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PART A – STUDY DESIGN & CORPUS CONSTRUCTION

Rationale: Data quality and study transparency are primary causes of rejection in top-tier peer review. These features convert a "tool run" into an auditable, reproducible study design. They must be implemented before analysis features to avoid invalidating outputs downstream.

A.1 Protocol Builder [NEW-H] [H 1]

A guided pre-analysis workspace that forces the user to define and record research questions, database selection rationale, inclusion/exclusion criteria, timespan, document types, search strings, and analytical plan before any bibliometric operation is run. Each field is versioned and forms part of the reproducibility bundle. Converts a "tool run" into an auditable study design — the absence of this step is a frequent reviewer objection in high-impact methods papers.
Tools: No major bibliometric platform provides this end-to-end; adjacent support is available via PRISMA workflows (prisma-statement.org) and Rayyan's review workspace.
[GAP] No bibliometric analysis tool implements this as a first-class feature.

A.2 Multi-Database Query Translation [NEW-H] [H 1]

Converts one master Boolean search strategy into syntactically equivalent queries for Scopus, Web of Science, OpenAlex, PubMed, and Lens, preserving field logic (title, abstract, keyword, MeSH terms) as far as each API supports. Reviewer criticism frequently begins with search incompleteness or database bias; automated translation makes multi-database coverage defensible.
Tools: No mainstream bibliometric tool provides cross-database query translation. Related ingestion breadth exists in Bibliometrix, VOSviewer, Lens.org, and Dimensions.
[GAP] High-differentiation feature; not implemented anywhere.

A.3 Multi-source database ingestion [PHASE 1]

Imports records from Web of Science (WoS), Scopus, PubMed, Dimensions, OpenAlex, Lens.org, Cochrane, and Google Scholar in their native export formats (BibTeX, RIS, CSV, TXT, XML). Each source retains its provenance tag throughout the entire pipeline, enabling source-level audit.
Tools: Bibliometrix, PyBibX, SciMAT, Sci2

A.4 AI-Assisted Screening with Human Audit Trail [NEW-H] [H 1]

Screens titles and abstracts using active machine-learning recommendations, while storing every inclusion/exclusion decision, the reason for exclusion, reviewer identity, and confidence signal. Accelerates corpus construction without sacrificing defensibility in peer review. The audit trail is exported as part of the reproducibility bundle.
Tools: Rayyan (AI-assisted screening, deduplication, exclusion reasons, collaboration, PRISMA-oriented workflows — production-ready).
[GAP] Not integrated into any major bibliometric analysis platform.

A.5 PRISMA Flow Diagram and Checklist Generator [NEW-H] [H 1]

Auto-builds the PRISMA 2020 flow diagram and a partially completed reporting checklist directly from the ingestion and screening logs, without manual entry. Many top-tier reviews now require PRISMA-style reporting; automation removes a common source of error and saves hours of preparation.
Tools: Rayyan (PRISMA flow creation); Bibliometrix (partial via workflow scripts). PRISMA standard: prisma-statement.org.
[GAP] Not fully automated in any major bibliometric tool.

A.6 Cross-database fusion and record merging [NEW-GS] [PHASE 1]

A normalization engine that ingests records from Scopus, WoS, Dimensions, and OpenAlex simultaneously, automatically detects field-tag conflicts (e.g., "AU" vs. "AF" vs. "author"), resolves them using configurable priority rules, and produces a single deduplicated master corpus. Deduplication uses fuzzy matching on the title, DOI, author list, and year, while preserving the source history and the merge rationale for every record.
Tools: VantagePoint, Bibliometrix (R) — partial; Rayyan for review workflows; SciMAT (Levenshtein-based preprocessing).
[GAP] No open-source tool performs complete field-tag conflict resolution with full provenance preservation.

A.7 Advanced entity disambiguation [NEW-GS] [PHASE 1]

Uses NLP pipelines and fuzzy string matching to merge author name variations (e.g., "J. Smith", "John Smith", "Smith, J.A.") and institutional aliases (e.g., "MIT", "Mass. Inst. of Tech.", "Massachusetts Institute of Technology") into canonical entities. Cross-references ORCID iDs and ROR (Research Organization Registry) for verified disambiguation. Essential before any collaboration or network analysis is reported in a high-tier publication.
Tools: VantagePoint, OpenRefine; Bibliometrix and SciMAT provide partial support. Full ORCID+ROR integration is absent in open-source tools.
[GAP] Complete NLP-based disambiguation with ORCID/ROR is not available in any major open-source bibliometric platform.

A.8 Context-aware stop-word and noise filtering [NEW-GS] [PHASE 1]

A domain-sensitive NLP filter that removes generic stop-words while preserving discipline-specific jargon. The stop-word list is automatically adapted to the detected research domain (e.g., "cell" is noise in bibliometrics but a key term in biology). Supports configurable allow-lists and deny-lists per project, stored in the reproducibility bundle.
Tools: VantagePoint, LHC (Leximancer/Human Coding tools); not natively available in Bibliometrix, VOSviewer, or CiteSpace.
[GAP] No open-source bibliometric tool provides domain-adaptive filtering.

A.9 Keyword cleaning and thesaurus merging [PHASE 1]

Identifies and unifies keyword synonyms, abbreviations, and plural/singular variants (e.g., "ML" = "machine learning") through configurable thesaurus files and semi-automated suggestions. Thesaurus is versioned and exportable.
Tools: VOSviewer (thesaurus file), SciMAT, Bibliometrix (partial)

A.10 Retraction and correction flagging [PHASE 1]

Cross-checks all corpus records against the Retraction Watch database and PubMed retraction notices; flags or excludes retracted articles before any metric computation to prevent propagation of invalid citations.
[GAP] Not implemented in any major bibliometric tool.

A.11 Corpus quality scoring dashboard [PHASE 1]

Flags records with missing abstract, DOI, or affiliation. Assigns a corpus quality score (0–100) based on completeness and recommends corrective actions (e.g., "32% of records lack abstracts; text mining reliability will be reduced") before proceeding. Score is logged in the reproducibility bundle.
[GAP] No current tool provides systematic scoring of corpus quality.

A.12 API-based live data retrieval [PHASE 1]

Direct connection to OpenAlex API, Dimensions API, PubMed Entrez, and CrossRef to build and update corpora programmatically without manual export steps. Enables scheduled refreshes and living systematic reviews.
Tools: PyBibX (OpenAlex, PubMed), Bibliometrix (OpenAlex, partial)

A.13 Full-text acquisition pipeline [PHASE 1]

Retrieves full-text PDFs or XML via open-access repositories (PubMed Central, Unpaywall, CORE) and extracts structured metadata and body text for downstream text mining and claim-level analysis. Tag each record with its OA status and available full-text version.
Tools: Lens.org (Unpaywall-enhanced OA discovery), Dimensions (full-text search indexing), Open Knowledge Maps (OA content highlighting).
[GAP] No major bibliometric tool integrates a complete full-text pipeline.

A.14 Reproducibility Bundle Export [NEW-H] [PHASE 1]

One-click export of the complete study package: search strategies, raw source files, cleaned corpus, thesauri, exclusion logs with reasons, parameter settings, software versions, normalization recipes, and figure/table generation scripts. Forms the supplementary material that enables replication by reviewers and readers.
Tools: Partial support via SciMAT reporting and Bibliometrix scriptable workflows, but not as a single integrated bundle.
[GAP] No tool produces a complete, self-contained reproducibility bundle.

PART B – TEMPORAL & FRONTIER DYNAMICS

Rationale: These features provide the quantitative proof that a topic is important NOW, justifying the manuscript's timeliness and motivating the research gap. They require valid Part A data.

B.1 Burst detection (Kleinberg's algorithm) [NEW-GS] [PHASE 2]

A 2-state Hidden Markov Model (HMM) that scans citation or keyword frequency time series and identifies sudden, statistically significant surges. Produces a burst score and burst period for each term or reference. Essential for labeling a topic as "Hot" in manuscript text. Publication in Scientometrics or JOI requires explicit evidence of a burst.
Tools: CiteSpace (primary), Bibliometrix (burstR function), Sci2 Tool

B.2 Sigma (σ) indicator [NEW-GS] [PHASE 2]

A composite metric computed as Burstiness × Centrality (σ = B × C). Identifies papers or keywords that simultaneously surge in popularity AND occupy strategic structural positions in the co-citation or co-occurrence network. High-σ items are designated as "landmark" or "transformative" works in manuscript narratives, supported by a single defensible score.
Tools: CiteSpace (exclusive native implementation)
[GAP] Not implemented in Bibliometrix, VOSviewer, PyBibX, or SciMAT.

B.3 Price's Index tracking [NEW-GS] [PHASE 2]

Calculates the percentage of all references in the corpus citing papers published within the last 5 years (Price's Index of obsolescence). Tracks this metric over time to measure whether the field is becoming more or less "current" — a standard indicator of field maturity and dynamism cited in Scientometrics methodology papers.
Tools: Bibliometrix computes partial support.
[GAP] Not available as an automated, tracked longitudinal metric.

B.4 Thematic evolution (Sankey / alluvial diagram) [NEW-GS] [PHASE 2]

Visualizes how research themes merge, split, or drift across defined time slices (e.g., 2010–2015 vs. 2016–2020 vs. 2021–2025). Each band in the Sankey diagram represents a thematic cluster; band width is proportional to publication volume. Supports the evidence-based "evolution" narrative, a defining feature of strong systematic bibliometric manuscripts.
Tools: Bibliometrix/biblioshiny (primary Sankey output); SciMAT (temporal analysis, strategic diagrams, overlapping and evolution maps); VOSviewer (overlay visualization for temporal developments).
[GAP] Not available in CiteSpace with equivalent resolution.

B.5 Temporal thematic evolution workspace [NEW-H] [H 2]

An interactive workspace for slicing the corpus across any user-defined time windows and displaying theme emergence, persistence, merging, splitting, and decline as comparable side-by-side or layered views. Extends B.4 with a configurable UI and exportable per-slice statistics. This is often where the paper's "contribution" section is built — the workspace enforces that the contribution is grounded in data.
Tools: SciMAT (time-slicing, temporal analysis); VOSviewer (overlay visualization for temporal maps).

B.6 Co-authorship network analysis [PHASE 2]

Builds and visualizes networks in which nodes are authors, institutions, or countries, and edges represent co-authored papers. Computes centrality metrics (degree, betweenness, closeness) and detects communities.
Tools: VOSviewer, Bibliometrix, CiteSpace, Gephi, Pajek

B.7 Co-citation analysis [PHASE 2]

Constructs networks of co-cited references or authors to reveal the intellectual foundations and seminal works of a research domain.
Tools: VOSviewer, CiteSpace, HistCite, Sci2, Bibliometrix

B.8 Bibliographic coupling analysis [PHASE 2]

Links documents sharing common references to cluster contemporary literature into active research streams and identify current research fronts. Preferred for detecting ongoing active work (vs. co-citation, which better maps historical foundations).
Tools: VOSviewer, Bibliometrix, CitNetExplorer

B.9 Keyword co-occurrence network [PHASE 2]

Builds networks of author keywords, index keywords, or title/abstract terms that co-appear in the same documents; clusters and visualizes thematic groups.
Tools: VOSviewer, CiteSpace, Bibliometrix, SciMAT, Gephi

B.10 Keyword co-word trend analysis (temporal) [PHASE 2]

Tracks the rise and decline of specific keywords or keyword clusters over time using sliding-window frequency plots and topic lifecycle models (emergence → growth → maturity → decline).
Tools: ScientoPy, CiteSpace, Bibliometrix (trendTopics function)

B.11 Technology lifecycle / S-curve analysis [PHASE 2]

Models the maturity stage of a technology or research topic by fitting S-curves (logistic or Gompertz functions) to cumulative publication or patent counts over time, producing a lifecycle phase classification used in tech-mining and innovation manuscripts.
Tools: ScientoPy (partial), VantagePoint (partial)

B.12 Multi-normalization engine [NEW-H] [H 2]

Let’s users compute and compare network results under multiple similarity or normalization measures: association strength, equivalence index, inclusion index, Jaccard, cosine, fractional counting, and full counting. Produce side-by-side comparison outputs showing whether major findings are stable across choices. Reviewers in top-tier journals routinely ask whether cluster findings are artifacts of a single normalization choice.
Tools: SciMAT (multiple normalization schemes); VOSviewer (fractional counting, advanced clustering/layout parameterization).

PART C – STRUCTURAL ANALYSIS & NOVELTY MINING

Rationale: High-impact research often bridges distinct knowledge communities or identifies structural gaps. These features identify novelty, boundary-spanning character, and research opportunities — the basis for the "original contribution" argument required by top-tier journals.

C.1 Structural variation analysis (SVA) [NEW-GS] [PHASE 3]

Measures the "structural shock" a new paper or cluster delivers to the existing co-citation or keyword co-occurrence network by computing the change in network topology (modularity Δ, betweenness redistribution) between the network with and without the target item. High SVA scores predict boundary-spanning or transformative work.
Tools: CiteSpace (primary SVA module)
[GAP] Not available in Bibliometrix, VOSviewer, PyBibX, or SciMAT.

C.2 Dual-map overlay analysis [NEW-GS] [PHASE 3]

Maps the relationship between "citing journals" (the research front) and "cited journals" (the intellectual base) on a standardized global map of science derived from journal clustering. Citation trajectories crossing discipline boundaries provide direct evidence of interdisciplinary knowledge transfer — a publishable finding, and a common figure type in top Scientometrics papers.
Tools: CiteSpace (primary dual-map module), VantagePoint (add-on)
[GAP] Not implemented in Bibliometrix, VOSviewer, or PyBibX.

C.3 Comparative science mapping by metadata segments [NEW-H] [H 2]

Builds separate conceptual maps for different corpus subpopulations —countries, institutions, journals, periods, author groups, or funding profiles — and overlays or compares them to reveal structural differences between communities. Supports richer theoretical claims than a single pooled map and is grounded in a 2024 Scientometrics methodology paper.
Tools: A 2024 Scientometrics paper (doi:10.1007/s11192-024-05161-6) introduces this approach; not yet a standard feature in mainstream bibliometric platforms.
[GAP] Not implemented as a standard feature in any bibliometric tool.

C.4 Structural gap discovery [NEW-GS] [PHASE 4]

An algorithm that detects "white spaces" in a co-citation or keyword network — clusters that are thematically proximate but lack citation links. These structural absences identify original research gaps that the manuscript can propose as future directions. Based on network null-model comparison and unexplained modularity gaps.
Tools: ResearchRabbit (visual gap suggestion); proposals in Journal of Informetrics literature.
[GAP] Research-level concept; no production-ready open-source tool.

C.5 Novelty / white-space detector [NEW-H] [H 3]

Identifies underexplored intersections between themes, methods, regions, or sectors by cross-tabulating metadata facets and measuring co-publication density relative to expected density from a null model. Directly feeds discussion sections, future research agendas, and grant positioning.
Tools: Lens.org (white-space and horizon-scanning language); Open Knowledge Maps (AI-based topic clustering highlighting neglected subtopics); ResearchRabbit (dynamic citation maps and recommendation-driven discovery). No mainstream bibliometric platform integrates this as a formal publication-oriented module.
[GAP] Not available as a formal, automated feature in any tool.

C.6 Visual discovery maps for gap finding [NEW-H] [H 3]

Clusters search results into intelligible topical areas so users can quickly see dominant zones and neglected subtopics in a visual landscape. Accelerates the transition from corpus inspection to the articulation of research gaps. Complements structural gap analysis (C.4) with visual accessibility for non-specialist readers and reviewers.
Tools: Open Knowledge Maps (AI-based knowledge maps with topic labels); ResearchRabbit (dynamic citation maps and recommendation-driven discovery).
[GAP] Not integrated into any mainstream bibliometric analysis platform.

C.7 Citation-network drill-down [NEW-H] [H 2]

Starts from a full citation network, then progressively drills into clusters, subclusters, and key documents. Supports precise historical and intellectual-structure narratives in the Introduction and Discussion of manuscripts.
Tools: CitNetExplorer (drill-down, expand, connected components, clusters, core publications, path analysis — production-ready).

C.8 Core publication and path analysis [NEW-H] [H 2]

Detects the most structurally central publications in the network and traces the shortest and longest knowledge-development paths, producing evidence for claims about seminal works, intellectual turning points, and field lineage.
Tools: CitNetExplorer (core publications, shortest/longest path algorithms); Bibliometrix (histNetwork); Pajek (main path analysis)

C.9 Main path analysis [PHASE 3]

Extracts the dominant citation pathway through a document network using Search Path Count (SPC) or Search Path Link Count (SPLC) algorithms, revealing the core developmental trajectory of a field and identifying the most intellectually influential papers.
Tools: Bibliometrix, Pajek

C.10 Historiograph / direct citation network [PHASE 3]

Constructs a chronological chain of direct citations among seminal papers to reveal the intellectual lineage of a field — the "family tree" of ideas. Essential for the Introduction section of top-tier manuscripts.
Tools: HistCite, CiteSpace (timeline view), Bibliometrix (histNetwork), CitNetExplorer

C.11 Betweenness and brokerage analysis [PHASE 3]

Identifies key bridging actors (authors, institutions, or topics) that connect otherwise disconnected communities in a collaboration or co-citation network — knowledge transfer agents and interdisciplinary gatekeepers.
Tools: Gephi, Pajek, Bibliometrix (partial)

C.12 Research community detection (modularity clustering) [PHASE 3]

Applies community detection algorithms (Louvain, Leiden, Walktrap) to identify groups of closely collaborating authors or thematically coherent documents within large networks.
Tools: Gephi (Louvain), VOSviewer (Smart Local Moving algorithm), Bibliometrix (cluster walktrap)

C.13 Thematic map (strategic diagram) [PHASE 3]

Position research themes on a 2×2 matrix defined by centrality (relevance to the field) and density (internal development). Classifies clusters as Motor Themes, Basic Themes, Niche Themes, or Emerging/Declining Themes —a standard framework appearing in nearly every Scientometrics bibliometric paper.
Tools: Bibliometrix/biblioshiny, SciMAT

C.14 Support/contrast citation layer [NEW-H] [H 3]

Classifies every citation in the corpus as supporting, contrasting, or mentioning, and exposes the exact citation context sentence. Enables separation of high-consensus claims from contested ones. A paper frequently cited as "contrasting" has a fundamentally different intellectual role than one cited as "supporting" — a dimension invisible to raw counts.
Tools: Scite.ai (Smart Citations — classification of citation intent/context, production-ready commercial); Semantic Scholar (citation intent classification, open API).
[**GAP**] Not integrated into any open-source bibliometric analysis platform.

C.15 Claim-level evidence mapping [NEW-H] [H 3]

For each major theme identified in the corpus, maps not only who cites whom, but also which empirical or conceptual claims are reinforced or disputed across publications. Supports deeper synthesis than standard co-citation maps and enables the Discussion section to characterize the state of evidence with precision.

Tools: Scite.ai provides the closest enabling layer; no standard bibliometric package fully integrates claim-level mapping.
[GAP] Not implemented as a production feature in any tool.

C.16 Research-to-technology transfer (Pub–Patent convergence) [PHASE 3]

Links scientific publications to citing patents (or vice versa) to measure and visualize the translation of basic research into applied technology. Traces knowledge flows from academia to industry using cross-database citation bridges.
Tools: Lens.org (reversible patent–scholarly joins); Dimensions (links publications, patents, grants, and clinical trials).
[GAP] No integrated open-source tool performs this end-to-end.

C.17 Grant–publication–clinical trial linkage [NEW-H] [H 3]

Traces how research funding produces outputs and how those outputs connect to patents or clinical trials. Enables causal and ecosystem narratives about the translational pipeline from funding through discovery to application — particularly powerful for health, engineering, and policy manuscripts.
Tools: Dimensions (explicit cross-source links among publications, grants, patents, researchers, and clinical trials — production-ready).
[GAP] Not integrated into any open-source bibliometric tool.

PART D – AI-DRIVEN INTERPRETATION & AUTHORING

Rationale: These features convert complex networks and statistical outputs into human-readable, publication-quality narratives, accelerating the path from analysis to submission. They depend on all prior parts producing clean, structured inputs. The key constraint is that every AI output must be grounded in specific, verifiable analytical results.

D.1 Generative cluster labeling (LLM-assisted) [NEW-GS] [PHASE 4]

Uses large language models (GPT-4o, LLaMA 3) to analyze the abstracts of papers within each thematic cluster and generate concise, accurate, human-like thematic labels — replacing manual interpretation with reproducible, prompt-engineered labeling. Labels include confidence scores and evidence of keyword support. Reduces the most time-intensive manual step in science mapping workflows.
Tools: CiteSpace 7.0 (integrated GPT labeling); Bibliometrix/Biblio AI (AI-assisted commentary, experimental); PyBibX (GPT integration)
[GAP] Not yet standardized or version-controlled in production releases.

D.2 Grounded AI interpretation of results [NEW-H] [H 4]

Generates narrative interpretations of tables, maps, and trends, but forces every paragraph to reference specific outputs, parameter settings, and cluster IDs. Saves time without turning the Results section into generic AI prose. Grounding prevents hallucinations and ensures that every claim is traceable to an analytical output.
Tools: Bibliometrix 5.0 / Biblio AI (AI-assisted commentary and synthesis for bibliometric outputs — production update).
[GAP] Full grounding with output-level citations not yet implemented.

D.3 Figure-to-text evidence linking [NEW-H] [H 4]

Every sentence in the AI-assisted narrative interpretation links back to the exact table, map, cluster, or statistic that justifies it. Creates a traceable chain from analytical output to manuscript sentence, reducing hallucination risk and improving reviewer trust. Often called "faithfulness" in the LLM evaluation literature.
Tools: No mainstream tool implements this tightly; the closest move is Biblio AI's AI commentary in Bibliometrix 5.0.
[**GAP**] High-differentiation feature; not implemented anywhere.

D.4 Methods Section Auto-Drafter [NEW-H] [H 4]

Drafts a manuscript-ready Methods section from the actual logged workflow: databases queried, retrieval dates, Boolean query strings, screening decisions, cleaning operations, matrices constructed, algorithms applied, normalization schemes used, and visualization parameters. Eliminates mismatch between what was done and what is reported — a major source of reviewer objections.
Tools: No major platform does this comprehensively; related structured workflows exist in Bibliometrix and PRISMA ecosystems.
[GAP] High-value differentiator; not implemented anywhere.

D.5 Reviewer-ready robustness appendix [NEW-H] [H 4]

Auto-generates sensitivity checks across co-occurrence thresholds, normalization choices, time windows, minimum citation counts, and alternative clustering algorithm settings. Produces a formatted appendix showing whether major findings — cluster structures, central nodes, key themes — are stable across reasonable analytical variations. Top-tier reviewers routinely request evidence of stability.
Tools: Partial support through parameterized tools (VOSviewer, SciMAT, CitNetExplorer), but not as an integrated robustness report.
[GAP] Not implemented as an automated integrated feature.

D.6 Dual-tool validation [NEW-H] [H 1]

Runs the same conceptual or structural analysis with two different algorithmic engines and presents a side-by-side comparison of cluster stability, theme consistency, and major findings. A 2025 methodology paper in Discover Applied Sciences explicitly recommends dual-tool analysis to improve interpretability and validation. The platform would automate the comparison rather than merely permitting it.
Tools: Literature discusses combined VOSviewer + biblioshiny/Bibliometrix workflows; no tool automates the comparison output.
[GAP] The comparison is documented in the literature but not automated.

D.7 BERTopic / neural topic modeling [PHASE 4]

Uses transformer-based sentence embeddings (SBERT) combined with UMAP dimensionality reduction and HDBSCAN density clustering to extract semantically coherent topics from abstracts. Produces more interpretable, context-sensitive topics than LDA, with an integrated dynamic mode for tracking temporal evolution. Useful as the primary topic model or as the comparison layer against LDA in a dual-model validation.
Tools: PyBibX (BERTopic integration); standalone Python (BERTopic library)
[GAP] Not natively integrated in Bibliometrix, VOSviewer, or CiteSpace GUI.

D.8 LDA topic modeling [PHASE 4]

Applies probabilistic Latent Dirichlet Allocation to discover latent topics across abstracts, providing an interpretable set of keywords per topic. Serves as the classical baseline against which BERTopic results can be compared for methodological validation.
Tools: PyBibX, Sci2 (via MALLET), standalone Python (gensim)

D.9 Dynamic topic modeling (topic evolution over time) [PHASE 4]

Tracks how vocabulary and prevalence of topics change across temporal windows, producing topic trajectories and evolutionary Sankey diagrams using sliding time-window BERTopic or dynamic LDA.
Tools: CiteSpace (time-slice analysis), standalone Python (BERTopic dynamic)
[GAP] Limited native support in major bibliometric GUI tools.

D.10 LLM-powered natural language corpus querying [PHASE 4]

Allows users to query the analyzed corpus in plain natural language (e.g., "Which institutions bridge medicine and computer science?", "What are the three most cited emerging topics since 2020?") through an integrated LLM interface that translates questions into analysis operations.
[GAP] Emerging research prototype; not in production in any tool.

D.11 Living review/update monitor [NEW-H] [H 3]

After the first publication, continuously monitors new papers matching the original search criteria, new citation clusters, and new evidence on key claims. Generates change-detection reports that justify a follow-up paper or a published update. Turns a one-off study into a continuing research program and dramatically increases the long-term return on one analysis.
Tools: ResearchRabbit (collections/monitoring); OpenAlex API ecosystem; VOSviewer (periodic re-runs with network comparison).
[GAP] Not integrated as a journal-production workflow in any tool.

D.12 Journal-framing assistant [NEW-H] [H 4]

Reframes the same analysis for different target publication outlets: methods-heavy Scientometrics journals, domain systematic reviews, innovation and policy venues, or management and strategy journals. Adjusts the framing of the Introduction, the emphasis of the Results, and the language of the Discussion to match the editorial expectations of the selected target. The difference between acceptance and rejection is often a matter of framing, not computational quality.
Tools: No mainstream bibliometric platform includes this feature.
[GAP] Novel; not implemented anywhere.

D.13 Automated narrative report generation [PHASE 4]

Generates a complete structured draft of the Methods and Results sections of a bibliometric manuscript, incorporating all computed statistics, network descriptions, cluster labels, and trend findings in natural-language paragraphs ready for researcher review and editing.
Tools: PyBibX (partial narrative summaries); Biblio AI (Bibliometrix 5.0)
[GAP] Full end-to-end manuscript drafting is not yet production-ready.

PART E – EXTENDED DOMAIN FEATURES

SECTION E.1 – PERFORMANCE ANALYSIS & BIBLIOMETRIC INDICATORS

[PHASE 1] E1.01 Publication output analysis

Computes annual publication counts, cumulative growth curves, and compound annual growth rates (CAGR) segmented by author, institution, country, journal, or keyword.
Tools: Bibliometrix/biblioshiny, PyBibX, ScientoPy, Sci2

[PHASE 1] E1.02 Citation impact metrics (h-index family)

Calculates total citations, average citations per document, h-index, g-index, hg-index, i10-index, and m-quotient for authors, journals, institutions, and countries.
Tools: Bibliometrix, Publish or Perish, PyBibX, Sci2

[PHASE 2] E1.03 Field- and context-normalized citation indicators [NEW-H]

Computes field-normalized citation scores (FNCS), category-normalized citation impact (CNCI), journal-normalized citation impact, and normalizations by country size, GDP, or research workforce — enabling fair cross-discipline and cross-country comparisons. A 2025 methodology paper explicitly recommends normalization with economic indicators.
Tools: SciVal (commercial), InCites (commercial); Dimensions provides partial data infrastructure.
[GAP] Not available in open-source bibliometric packages.

[PHASE 1] E1.04 Journal impact and ranking analysis

Retrieves and displays JIF, CiteScore, SJR, SNIP, and quartile rankings for all source journals in the corpus.
Tools: Bibliometrix (partial via Scopus data), PyBibX

[PHASE 1] E1.05 Bradford's Law/core journal analysis

Identifies the core set of journals concentrating most publications on a topic (Bradford zones), guiding submission strategy.
Tools: Bibliometrix, ScientoPy

[PHASE 1] E1.06 Lotka's Law/author productivity distribution

Models the inverse square law of author productivity.
Tools: Bibliometrix

[PHASE 1] E1.07 Zipf's Law/keyword frequency analysis

Analyzes term frequency distributions in titles, abstracts, and author keywords.
Tools: Bibliometrix, VOSviewer, ScientoPy

[PHASE 1] E1.08 Collaboration index and internationalization metrics

Computes co-authorship rates, ICI, single-country publication ratio, and mean authors per document.
Tools: Bibliometrix, PyBibX, Sci2

[PHASE 3] E1.09 Open access and altmetrics integration

Reports open-access status and retrieves Altmetric and Unpaywall scores.
Tools: Lens.org (Unpaywall-enhanced OA), PyBibX (partial)
[GAP] Not integrated in most open-source tools.

[PHASE 4] E1.10 SDG (Sustainable Development Goals) classification

Tags publications with relevant UN SDGs using keyword classifiers or trained ML models.
[GAP] Partial support in Dimensions and OpenAlex.

SECTION E.2 – VISUALIZATION

[PHASE 2] E2.01 Interactive network visualization

Renders co-authorship, co-citation, and keyword networks as interactive, zoomable graphs with community color-coding; exports to SVG, PDF, HTML.
Tools: VOSviewer, Gephi, CiteSpace, Bibliometrix (igraph-based)

[PHASE 2] E2.02 Density/heat map overlay for networks

Overlays a kernel density estimate on the network map to highlight areas of high publication concentration.
Tools: VOSviewer (density view)

[PHASE 2] E2.03 Temporal/timeline visualizations

Displays clusters, keywords, or authors along a time axis.
Tools: CiteSpace (time-zone/time-slice), HistCite, CitNetExplorer

[PHASE 2] E2.04 Collaboration world map

Geospatial visualization of international collaborations with proportional arc lines between collaborating countries.
Tools: Bibliometrix/biblioshiny, PyBibX, VOSviewer

[PHASE 1] E2.05 Interactive dashboard / no-code GUI

Web-based GUI for point-and-click analysis without programming.
Tools: biblioshiny (R Shiny), VOSviewer (desktop GUI)

[PHASE 3] E2.06 Full-text-aware term mining [NEW-H]

Moves beyond title/abstract/keywords by extracting relevant terms from full text when licensing allows, then algorithmically separates relevant from non-relevant terms to improve conceptual sensitivity.
Tools: VOSviewer (relevant/non-relevant term distinction, term maps); Dimensions (full-text search indexing).

SECTION E.3 – REPRODUCIBILITY & WORKFLOW MANAGEMENT

[PHASE 1] E3.01 Versioned corpus snapshots

Saves time-stamped, immutable snapshots of the corpus at each analysis stage for full reproducibility and traceability.
[GAP] Not implemented in any major bibliometric tool.

[PHASE 1] E3.02 Provenance and methodological logging

Automatically logs all analysis parameters (database version, retrieval date, threshold settings, normalization method, clustering algorithm) and exports a structured Methods section in natural language.
[GAP] No current tool generates a complete structured methods log.

[PHASE 1] E3.03 R/Python scripting API with reproducible workflows

Exposes all analysis functions through a well-documented API, enabling fully scripted, reproducible workflows shareable as supplementary materials.
Tools: Bibliometrix (R), PyBibX (Python)

[PHASE 2] E3.04 Multi-tool interoperability (export/import bridges)

Exports outputs in formats for complementary tools: GraphML, GML, Pajek .net, VOSviewer map files, GEXF, Excel, CSV, BibTeX.
Tools: Bibliometrix, SciMAT, VOSviewer (partial)

[PHASE 4] E3.05 Cloud-based collaborative workspace

Multi-user cloud platform for shared corpus access, parallel analysis, and co-authoring of analysis reports.
[GAP] Not available in any current open-source bibliometric tool.

SECTION E.4 – TECH MINING & TECHNOLOGY INTELLIGENCE

[PHASE 3] E4.01 Patent landscape mapping

Visual maps of the patent space using IPC/CPC codes, assignee networks, citation links, and abstract text.
Tools: VantagePoint, PatSeer, AcclaimIP, PatentInspector

[PHASE 3] E4.02 Technology opportunity analysis / white spot detection

Identifies underdeveloped technology facet combinations by cross-tabulating patent metadata.
Tools: VantagePoint; Porter & Cunningham (2005) framework
[GAP] Not implemented in open-source academic tools.

[PHASE 4] E4.03 Technology forecasting / emerging topic prediction

Machine learning models (Random Forest, AdaBoost, GCN) trained on bibliometric features to predict future mainstream topics.
[GAP] Active research; no production open-source implementation.

[PHASE 3] E4.04 Competitive intelligence profiling

Profiles organizations by publication and patent portfolios vs. peers.
Tools: SciVal (commercial), InCites (commercial), VantagePoint (partial)
[GAP] Not available in open-source tools.

[PHASE 3] E4.05 Innovation pathway analysis (FIP approach)

Reconstructs the developmental trajectory of a technology combining publication trends, patent analysis, and expert knowledge.
Tools: VantagePoint (partial); Porter et al. (2015) framework
[GAP] No fully automated open-source implementation.

SECTION E.5 – PATENT-SPECIFIC TECH-MINING FEATURES

[PHASE 1] E5.01 Patent database integration

Imports patent records from USPTO, EPO, PatBase, Lens.org, Google Patents with IPC/CPC codes as first-class metadata.
Tools: VantagePoint, PatentInspector, Lens.org
[GAP] Not integrated into general bibliometric platforms.

[PHASE 2] E5.02 IPC/CPC classification-based clustering

Groups patents into technology domains by classification code.
Tools: VantagePoint, PatentInspector, AcclaimIP, Lens.org

[PHASE 3] E5.03 Assignee and inventor network analysis

Builds co-inventor and co-assignee networks to map innovation ecosystems.
Tools: VantagePoint, PatSeer, PatentInspector

[PHASE 2] E5.04 Patent family analysis

Groups filings across jurisdictions to avoid double-counting.
Tools: PatBase, Minesoft Origin, VantagePoint

[PHASE 2] E5.05 Forward/backward citation analysis for patents

Traces citing and cited patents to assess downstream impact and prior art.
Tools: Lens.org (PatCite), AcclaimIP, PatSeer

[PHASE 3] E5.06 Technology white spot detection (patent gap analysis)

Identifies technology facet combinations lacking patent protection.
Tools: VantagePoint (partial)
[GAP] Not available in open-source platforms.

[PHASE 4] E5.07 NLP-based claim segmentation and feature extraction

Parses patent claim language to extract functional features and solutions.
Tools: Linguamatics I2E, PatentInspector (partial)

SECTION E.6 – AI-AUGMENTED ANALYSIS (EXTENDED)

[PHASE 4] E6.01 Named entity recognition (NER) for scientific text

Extracts chemicals, diseases, organisms, methods, datasets, and institutions from titles/abstracts using trained NER models.
Tools: Linguamatics I2E (commercial), standalone NLP (spaCy, SciSpacy)
[GAP] Not integrated into any major open-source bibliometric platform.

[PHASE 4] E6.02 Knowledge graph construction from corpus

Builds a structured RDF/OWL knowledge graph linking entities through typed relationships, enabling semantic querying.
[GAP] Research prototype (SciKGraph); not in production tools.

[PHASE 4] E6.03 Research topic recommendation for scholars

Suggests underexplored high-potential research topics based on a scholar's publication history and emerging field signals.
[GAP] Research prototype (LLM-TRec); not in any GUI tool.

[PHASE 4] E6.04 Bias detection in citation and collaboration networks

Identifies systematic biases related to gender, geography, language, or institutional prestige using statistical tests and ML classifiers.
[GAP] Emerging research; no standardized tool.

[PHASE 4] E6.05 Real-time corpus monitoring and alerting

Continuously monitors database queries and alerts users to new matching publications or patents.
Tools: Minesoft Origin (patents), Google Scholar Alerts
[GAP] Not integrated into open-source bibliometric tools.

[PHASE 4] E6.06 Multi-modal analysis (figures, tables, data)

Extracts and analyzes structured data from full-text PDFs using multimodal LLMs to complement metadata-level analysis.
[GAP] No bibliometric tool implements this.

GOLD STANDARD FEATURE COVERAGE MATRIX

Phase completion — estimated percentage of features in each implementation phase currently supported at the production level:

Tool	Phase 1	Phase 2	Phase 3	Phase 4	Overall
Bibliometrix	75%	70%	60%	35%	62%
CiteSpace	45%	85%	60%	40%	58%
PyBibX	70%	55%	40%	55%	56%
VOSviewer	40%	65%	45%	10%	44%
SciMAT	60%	50%	50%	10%	45%
Rayyan	80%	5%	5%	5%	30%
CitNetExplorer	30%	55%	60%	5%	40%
Lens.org	50%	30%	50%	10%	38%
Scite.ai	5%	5%	5%	70%	18%
Dimensions	55%	30%	40%	15%	38%
VantagePoint	55%	45%	65%	20%	49%
Gold Standard	100%	100%	100%	100%	100%

Part coverage (★ = 5-point scale):

Part / Section	Bibliometrix	VOSviewer	CiteSpace	PyBibX	SciMAT
A. Study Design & Corpus	★★★★☆	★★☆☆☆	★★★☆☆	★★★★☆	★★★☆☆
B. Temporal & Frontier	★★★★☆	★★★★☆	★★★★★	★★★☆☆	★★★☆☆
C. Structural & Novelty	★★★★☆	★★★★☆	★★★★☆	★★★☆☆	★★★★☆
D. AI Interpretation & Authoring	★★★☆☆	★☆☆☆☆	★★★☆☆	★★★★☆	★☆☆☆☆
E1. Performance Indicators	★★★★★	★★☆☆☆	★★☆☆☆	★★★★☆	★★★☆☆
E2. Visualization	★★★★☆	★★★★★	★★★★☆	★★★☆☆	★★★☆☆
E3. Reproducibility	★★★★☆	★★☆☆☆	★★☆☆☆	★★★☆☆	★★☆☆☆
E4. Tech Mining	★☆☆☆☆	★☆☆☆☆	★☆☆☆☆	★★☆☆☆	★☆☆☆☆
E5. Patent Analysis	★☆☆☆☆	★☆☆☆☆	★☆☆☆☆	★☆☆☆☆	★☆☆☆☆
E6. AI-Augmented (Extended)	★★☆☆☆	★☆☆☆☆	★★☆☆☆	★★★☆☆	★☆☆☆☆

★★★★★ Fully implemented ★★★★☆ Mostly implemented ★★★☆☆ Partially implemented
★★☆☆☆ Minimal support ★☆☆☆☆ Not implemented

RECOMMENDED TOOL STACK FOR TOP-TIER JOURNAL PUBLICATION

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MINIMUM VIABLE STACK (covers Phases 1–4 for standard bibliometric study):
  Bibliometrix/Biblioshiny  →  Phase 1 data engineering, Phase 2 trends,
                                Phase 3 structure, reproducibility API,
                                Biblio AI for narrative (Phase 4)
  CiteSpace                 →  Phase 2 burst detection, sigma indicator;
                                Phase 3 SVA and dual-map overlay
  PyBibX                    →  Phase 4 BERTopic, LDA, LLM cluster labeling,
                                narrative output generation
  Scite.ai                  →  Phase 4 sentiment/support citation analysis
  Rayyan                    →  Phase 1 AI-assisted screening, PRISMA flow,
                                deduplication with audit trail

EXTENDED STACK (adds novelty mining, tech mining, patent analysis):
  VantagePoint / OpenRefine →  Phase 1 advanced entity disambiguation
  Lens.org                  →  Phase 3 Pub–Patent convergence linkage,
                                OA full-text, patent–paper joins
  CitNetExplorer            →  Phase 3 citation drill-down, core publications,
                                path analysis
  ResearchRabbit /
  Open Knowledge Maps       →  Phase 3 visual gap finding and novelty detection
  Dimensions                →  Phase 3 grant–publication–trial linkage
  PatSeer / AcclaimIP       →  Phase 2–3 patent landscape and gap analysis
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PRIORITIZED IMPLEMENTATION ROADMAP (BUILD ORDER)

BUILD FIRST — Submission-grade methodological rigor

1.	Protocol Builder	(A.1)
2.	AI-Assisted Screening + Audit Trail	(A.4)
3.	PRISMA Flow Diagram and Checklist Generator	(A.5)
4.	Reproducibility Bundle Export	(A.14)
5.	Dual-Tool Validation	(D.6)
6.	Methods Section Auto-Drafter	(D.4)
7.	Multi-Database Query Translation	(A.2)
8.	Cross-database fusion with full provenance	(A.6)

BUILD SECOND — Deeper analytical power

9.	Sigma (σ) indicator in open-source	(B.2)
10.	Structural variation analysis (SVA)	(C.1)
11.	Dual-map overlay analysis	(C.2)
12.	Comparative science mapping by metadata segments	(C.3)
13.	Citation drill-down + core/path analysis	(C.7, C.8)
14.	Temporal thematic evolution workspace	(B.5)
15.	Multi-normalization engine + robustness appendix	(B.12, D.5)
16.	Field- and context-normalized performance indicators	(E1.03)
17.	Patent / grant / trial linkage	(C.16, C.17)

BUILD THIRD — Novelty generation and manuscript production

18.	Support/contrast citation layer (open-source)	(C.14)
19.	Claim-level evidence mapping	(C.15)
20.	Novelty / white-space detector	(C.5)
21.	Visual discovery maps for gap finding	(C.6)
22.	Grounded AI interpretation + figure-to-text linking	(D.2, D.3)
23.	Journal-framing assistant	(D.12)
24.	Living review / update monitor	(D.11)
25.	BERTopic integrated in GUI platform	(D.7)

CRITICAL GAPS BY STRATEGIC PRIORITY

Rank	Feature	Part	Impact
1	Protocol Builder (first-class feature)	A.1	★★★★★
2	Methods Section Auto-Drafter	D.4	★★★★★
3	Sigma (σ) indicator in open-source	B.2	★★★★★
4	Structural variation analysis (SVA)	C.1	★★★★★
5	Dual-map overlay analysis	C.2	★★★★★
6	Support/contrast citation (open-source)	C.14	★★★★★
7	Structural gap discovery algorithm	C.4	★★★★★
8	Figure-to-text evidence linking	D.3	★★★★★
9	Journal-framing assistant	D.12	★★★★★
10	Multi-database query translation	A.2	★★★★☆
11	Full cross-database fusion with tag resolution	A.6	★★★★☆
12	Reproducibility bundle export (automated)	A.14	★★★★☆
13	Price's Index as automated tracked metric	B.3	★★★★☆
14	Comparative science mapping by segment	C.3	★★★★☆
15	BERTopic integrated in GUI bibliometric tools	D.7	★★★★☆
16	Reviewer-ready robustness appendix	D.5	★★★★☆
17	Dual-tool validation (automated output)	D.6	★★★★☆
18	Living review / update monitor	D.11	★★★★☆
19	Retraction flagging from Retraction Watch	A.10	★★★☆☆
20	Corpus quality scoring dashboard	A.11	★★★☆☆
21	Normalized citation indicators (open-source)	E1.03	★★★☆☆
22	Pub–Patent convergence linkage (open-source)	C.16	★★★☆☆
23	Grant–publication–trial linkage	C.17	★★★☆☆
24	Claim-level evidence mapping	C.15	★★★☆☆
25	Novelty / white-space detector (formal module)	C.5	★★★☆☆

REFERENCES (KEY PAPERS & SOURCES)

Aria, M. & Cuccurullo, C. (2017). bibliometrix: An R-tool for comprehensive science mapping analysis. Journal of Informetrics, 11(4), 959–975.

Aria, M. (2024). Biblio AI — AI-assisted bibliometric commentary and synthesis. Bibliometrix 5.0. bibliometrix.org.

Chen, C. (2006). CiteSpace II: Detecting and visualizing emerging trends and transient patterns in scientific literature. JASIST, 57(3), 359–377.

Cobo, M.J. et al. (2011). Science mapping software tools: review, analysis, and cooperative study among tools. JASIST, 62(7), 1382–1402.

Donthu, N. et al. (2021). How to conduct a bibliometric analysis: An overview and guidelines. Journal of Business Research, 133, 285–296.

Grootendorst, M. (2022). BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv:2203.05794.

Ibrahim, A. et al. (2025). Innovative methodological pillars for bibliometric studies: AI screening, data normalization and dual-tool analysis. Discover Applied Sciences. doi:10.1007/s42452-025-07653-3.

Kleinberg, J. (2002). Bursty and hierarchical structure in streams. ACM SIGKDD Explorations Newsletter, 5(2), 91–101.

Moral-Muñoz, J.A. et al. (2024). Comparative science mapping: a novel conceptual structure analysis with metadata. Scientometrics. doi:10.1007/s11192-024-05161-6.

Page, M.J. et al. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. BMJ, 372:n71. prisma-statement.org.

Pereira, V. et al. (2025). PyBibX: A Python Library for Bibliometric and Scientometric Analysis Powered with AI Tools. Data Technologies and Applications, 59(2), 302–337.

Porter, A.L. & Cunningham, S.W. (2005). Tech Mining: Exploiting New Technologies for Competitive Advantage. Wiley.

Porter, A.L., Zhang, Y. & Newman, N.C. (2024). Tech mining: a revisit and navigation. Frontiers in Research Metrics and Analytics, 9:1364053.

Thelwall, M. et al. (2025). Research quality evaluation by AI in the era of large language models. Scientometrics. doi:10.1007/s11192-025-05361-8.

Tomaszewski, R. (2023). Visibility, impact, and applications of bibliometric software tools through citation analysis. Scientometrics, 128, 4007–4028.

Van Eck, N.J. & Waltman, L. (2010). Software survey: VOSviewer, a computer program for bibliometric mapping. Scientometrics, 84(2), 523–538.

SUMMARY

Total features documented: 91 (across 4 parts + 6 extended sections)
[NEW-GS] features from Gold Standard phased analysis: 12
[NEW-H] features from Horizon competitive-advantage analysis: 17
[GAP] features with no production open-source implementation: 34
Recommended minimum tool stack for top-tier publication: Bibliometrix + CiteSpace + PyBibX + Scite.ai + Rayyan